Jack Ford

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EDUCATION

The University of California, Davis **Davis, CA**
Master of Science in Computer Science *Expected June 2027*
Graduate Student Researcher - Data Enclaves for Secure Computing
Supervisor: Dr. Venkatesh Akella

Relevant Courses: Computer and Information Security, Operating Systems, Computer Architecture, Analysis of Software Artifacts

The University of Kansas **Lawrence, KS**
Bachelor of Science in Computer Science *May 18, 2025*
Overall GPA 3.95/4.0 | Credits: 123
Honors Student
Departmental Honors
Cybersecurity Undergraduate Certificate

Relevant Courses: Software Reverse Engineering, Introduction to Operating Systems (Gained exposure to GDB), Introduction to Information and Computer Security, Cyber Defense (gained exposure to Wireshark, NMAP, Snort, Security Onion, Metasploit), Introduction to Communication Networks, Programming Languages, Introduction to Database Systems, Introduction to Machine Learning, Embedded Systems, Computer Architecture, Programming Language Paradigms, Data Structures and Algorithms, Software Engineering 1 (gained exposure to Docker), Programming 1: Honors, Programming 2, Discrete Structures, Probability and Statistics, Introduction to the Theory of Computing, Calculus 1, Calculus 2, Calculus 3, Introduction to Digital Logic: Honors, Software Engineering II, Computer Science & Interdisciplinary Computing Capstone, Computer Forensics, Advanced Software Security Evaluation, Mobile Security, Compiler Construction

INTERNSHIPS

Oak Ridge National Laboratory **Oak Ridge, TN 37830**
Cyber Security Researcher Intern *June 02, 2025 – August 08, 2025*
Mentor: Curtis Taylor, (865) 241-5473

- Configured and deployed a 5G Stand-Alone Network using a Software Defined Radio
- Analyzed 5G Network radio frequency signals towards development of a testbed of 5G SDR Networks in a manufacturing environment

Air Force Civilian Service **Edwards Air Force Base, CA 93524**
Cyber Security Analyst Intern *May 20, 2024 – August 09, 2024*
Supervisor: Jason Schalow, (661) 277-9525

- Analyzed and developed techniques for de-obfuscating obfuscation methods
- Conducted a penetration test on a local machine, presenting findings and methodology to supervisors.
- Worked on mission-critical tasks, including Security Information and Event Management (SIEM) operations within the Security Operation Center (SOC) and performing malware analysis in a dedicated malware lab
- Regularly taught and presented cybersecurity concepts to peers, superiors, and students, including educational sessions for elementary students.

ACTIVITIES

Cybersecurity Club (cowsay) *August 2025 – Present*
Supervisor: Matthew Bishop, (530) 752-8060

- Managed the CTF team for the club and chose weekly CTF events for club members to participate in

- Competed in CTF events/tournaments and hacking competitions alongside a group of my peers
    - SunshineCTF
        - Practiced reverse engineering and writeup skills
    - Securinets CTF Quals 2025
    - Hack.lu CTF 2025
    - M0leCon CTF 2026 Teaser
    - RSTCON 2025 CTF
    - Buckeye CTF 2025
    - CPTC West Regional
        - Travelled to Stanford University and participated in a penetration testing competition as part of a team of six
        - Developed skills in web exploitation as well as documentation and teamwork
- Learned about current events and recent advancements in the field of Cyber Security
- Attended workshops and other networking events in pursuit of greater knowledge in the cybersecurity field

Information Security Club (Jayhackers) *March 2023 – May 2025*
Supervisor: Bo Luo, (785) 864-7749

- Competed in CTF events/tournaments and hacking competitions alongside a group of my peers
    - PicoCTF 2023 (Gained exposure to Ghidra)
    - SekaiCTF 2023
    - BuckeyeCTF
        - Practiced reverse engineering and binary exploitation concepts
    - USCCTF
        - 15th overall (individual)
        - Practiced reverse engineering and binary exploitation concepts
        - Practiced web exploitation, cryptography, forensics, and OSINT
    - 1337UP LIVE CTF
        - Practiced reverse engineering and binary exploitation concepts
    - Best Cyber Warrior X CTF Competition
        - 1st collegiate division (team)
        - 9th overall (team)
        - Practiced reverse engineering and binary exploitation concepts
    - NCL 2025
        - 7th overall out of over 450 teams
        - Practiced reverse engineering, networking, cryptographic, and analysis techniques
    - MetaCTF
        - Participate in monthly Flash CTFs
        - Solve challenges under the Reverse Engineering and Binary Exploitation categories
- Learned about current events and recent advancements in the field of Cyber Security

KU Men's Club Volleyball *August 2021 – May 2025*
President *April 2022 – May 2025*
Supervisor: Howard Graham, (785) 864-4277

- Lead group of about 30 peers
- Organize practices, travel arrangements, volunteer events, and tournaments
- Manage finances and dues working alongside the KU Sports Club organization to maintain good standing of the club
- Coordinate with the National Collegiate Volleyball Federation (NCVF) sending multiple KU teams to regular season tournaments (crossovers, regionals, conference) and national competitions

UC-Davis Men's Club Volleyball *September 2025 – Present*
Member *September 2025 – Present*

- Travelled as part of a team to tournaments and games
- Competed at a high level against other schools
- Developed relationships with peers in different academic fields

Upsilon Pi Epsilon (Computer Science Honor Society) *September 2023 – May 2025*
Supervisor: Drew Davidson, (913) 321-3143

Tau Beta Pi (Engineering Honors Society) *January 2024 – May 2025*
Supervisor: Lorin Maletsky, (785) 864-2985

SCHOLARSHIPS & ACHIEVEMENTS

- The CyberCorps Scholarship for Service (SFS) Program – 2-year scholarship + stipend and benefits (June 2023-May 2025)
- KU Excellence Award Recipient – $64,000 total for four years
- Garmin Award Recipient – $2,000 for 2022-23 school year
- Disney Scholars Award Recipient – $5,000 per year
- NMSQT National Rural and Small Town Recognition Program Award Recipient

RELEVANT PROJECTS

Github: https://github.com/shortestllama

- Obfuscation analysis: C/C++, x86-64
    - Solved obfuscated crackmes and ctfs from "crackmes.one" and "picoctf.org"
    - Worked on methods such as packers, anti-debug, anti-disassembly, self-modifying code, and encryption
- Packing/compression algorithms analysis: C/C++, x86-64
    - Investigated and explored UPX along with different compression techniques to discover their effectiveness
    - Successfully earned Departmental Honors following a presentation in front of a panel of three department faculty members
- Pic-A-Pass: Python
    - Worked as part of a team to develop a functional password manager that uses steganography as a unique component
    - Aided in the development of the program to ensure passwords were stored securely on each users' local device using password hashing and steganography
- Quash: C++
    - Terminal command line emulator based off of bash
    - Used and developed skills with operating system control execution
- Rustbucket: Rust
    - Utility with potential security exploits for security researchers and students alike
    - Developed in a weekend at HackKU25 to gain experience with rust
- Ro-Board: Python/Choregraphe
    - Robot that plays Connect Four against you
    - Developed robotic, kinematic, and networking skills
- Snake: Scratch
    - Developed a normal snake game in a block programming language
    - Learned new programming techniques from the limitations of the block programming language such as timing and object execution control
- Advanced Pong: Python3 (Pygame)
    - Traditional 8-bit Pong game made with the pygame library and OOP
    - The twist: upgrades randomly appear on the screen throughout the game to mess your opponent up if the ball goes through it
        - E.g. ball speeds up, ball slows down, ball splits into 3 paths, etc.
- Horde Defense: Python3 (Pygame)
    - 2D Black Ops Zombies-like game with rounds and increasing difficulty made with pygame library and OOP
        - 4 playable characters, each with their own unique weapon

SUMMARY OF SKILLS

First year Master's student at the University of California, Davis majoring in Computer Science and alumni of the University of Kansas with a Bachelor's degree in Computer Science focused on developing and expanding my knowledge and skills with a career in cybersecurity.

- Broadened my skills in cybersecurity as a Cyber Security Researcher Intern at Oak Ridge National Laboratory through a 10-week research and development project over the summer
- Gained practical experience as a United States Air Force Civilian Cyber Security Analyst Intern through the CyberCorps® Scholarship for Service Program
- Demonstrated abilities in problem-solving, pattern recognition, and building effective solutions for complex challenges
- Proficient in identifying security vulnerabilities, enhancing system defenses, and re-engineering processes for improved cybersecurity
- Passionate about contributing to government efforts in protecting critical infrastructure and national security, with a strong commitment to continuous learning and skill development
- Strong coding skills in multiple languages and experience across diverse operating systems:
  - Languages: Common CISC Assembly Languages, C/C++, Python, ARM, MIPS, Java, JavaScript, SQL, HTML, Markdown
  - Operating Systems: Linux (Kali, Ubuntu), Windows 11, macOS
- Other skills:
  - Languages: English (Native), German (Intermediate)
  - Technical: C/C++ – Proficient, x86-64 – Proficient, Linux – Proficient, Python – Proficient, Windows – Intermediate, SQL – Intermediate, Java – Intermediate, JavaScript – Intermediate, HTML – Intermediate, Advanced Mathematics – Proficient, iOS – Intermediate, Mac OS – Intermediate, Windows – Intermediate, Microsoft Office (Excel, Word, PowerPoint) – Intermediate, Social Media (YouTube, TikTok, Snapchat, Twitter, Instagram, Facebook, Reddit, Discord, Twitch) – Proficient

EMPLOYMENT REQUIREMENTS

Citizenship: American-born US Citizen
Veteran's Preference: No
Security Clearance: DoD Secret
Registered for Selective Service: Yes
Location Preferences: Flexible with a focus on California, Illinois, Washington D.C.

ADDITIONAL WORK EXPERIENCE

Lawrence Landsharks — **4120 Clinton Pkwy Frontage, Lawrence, KS 66047**
Assistant Volleyball Coach — *October 2022 – March 2023*
Supervisor: Kalissa McAtee, (785) 550-4222 — 18 hours/week, $25/practice, $50/tournament

- Helped oversee the execution of practice plans for 2 teams of approximately 10 high school aged girls
- Communicated with high schoolers, parents, officials, and other coaches on problem solving ideas
- Developed my public speaking abilities in front of large groups of people

Ambler Student Recreation and Fitness Center — **1740 Watkins Center Drive, Lawrence, KS 66045**
Weight Room Specialist — *June 2022 – November 2022*
Supervisor: Miranda Kolenda, (785) 864-1822 — 10 hours/week, $10/hour

- Monitored and cleaned a weightlifting facility with upwards of 300 students at any one time
- Rounded up towels around the facility to have washed and cleaned

Walmart — **2300 US-34, Oswego, IL 60543; 550 Congressional Drive, Lawrence, KS 66049**
Personal Shopper; Stocker — *June 2020 – September 2020; July 2022 – August 2022*

- Processed the storing, sorting, and distribution of upwards of 300 online grocery orders each day
- Coordinated 20 deliveries per hour to the customer's satisfaction in a fast-paced working environment alongside team members

Chicagoland Pool Management Group — **1612 Ogden Ave Suite 101, Lisle, IL 60532**
Lifeguard — *July 2018 – September 2018; June 2019 – September 2019*
Supervisor: Leslie Clark, (630) 388-8799 — 24 hours/week, $9/hour

- Facilitated a protective environment by conducting required safety protocols for 300 daily attendants
- Maintained the facilities of the job site throughout the day